

Experiment 15:

AI-based Traffic Prediction for 5G Networks

Objective 1:

To simulate 5G network traffic data and visualize the traffic pattern.

CODE :

```
clc;

clear;

close all;

% Simulated 5G Network Traffic (Mbps)

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Plot the graph

figure;

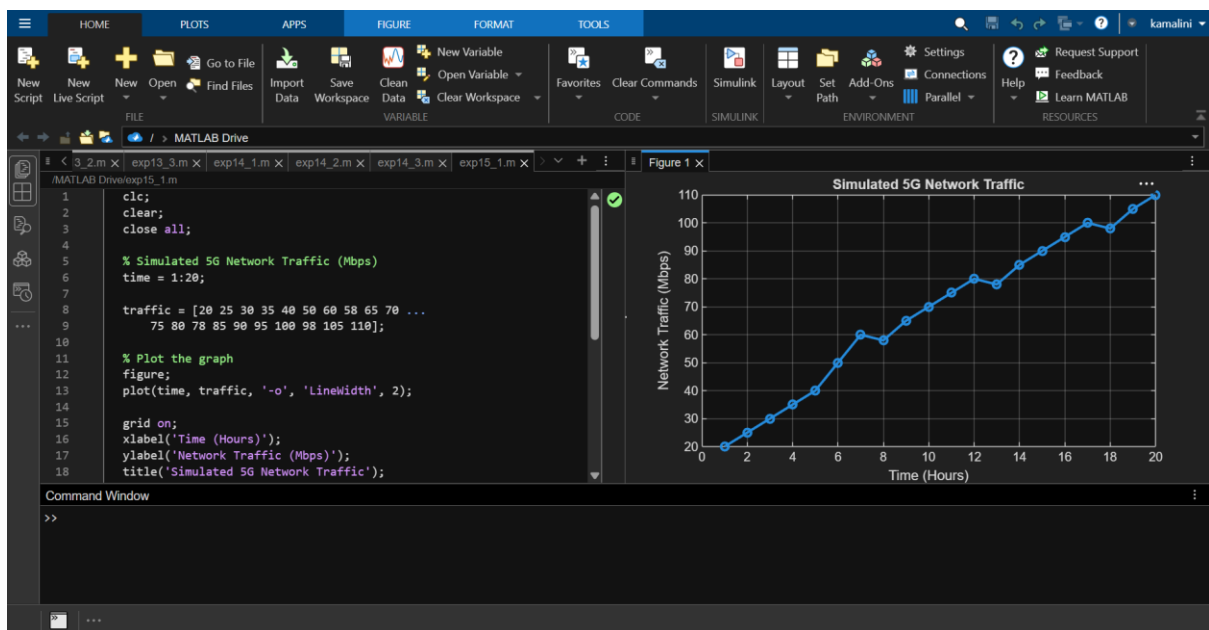
plot(time, traffic, '-o', 'LineWidth', 2)

grid on;

xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('Simulated 5G Network Traffic');
```



Objective 2:

To predict future 5G network traffic using Machine Learning (Linear Regression).

CODE :

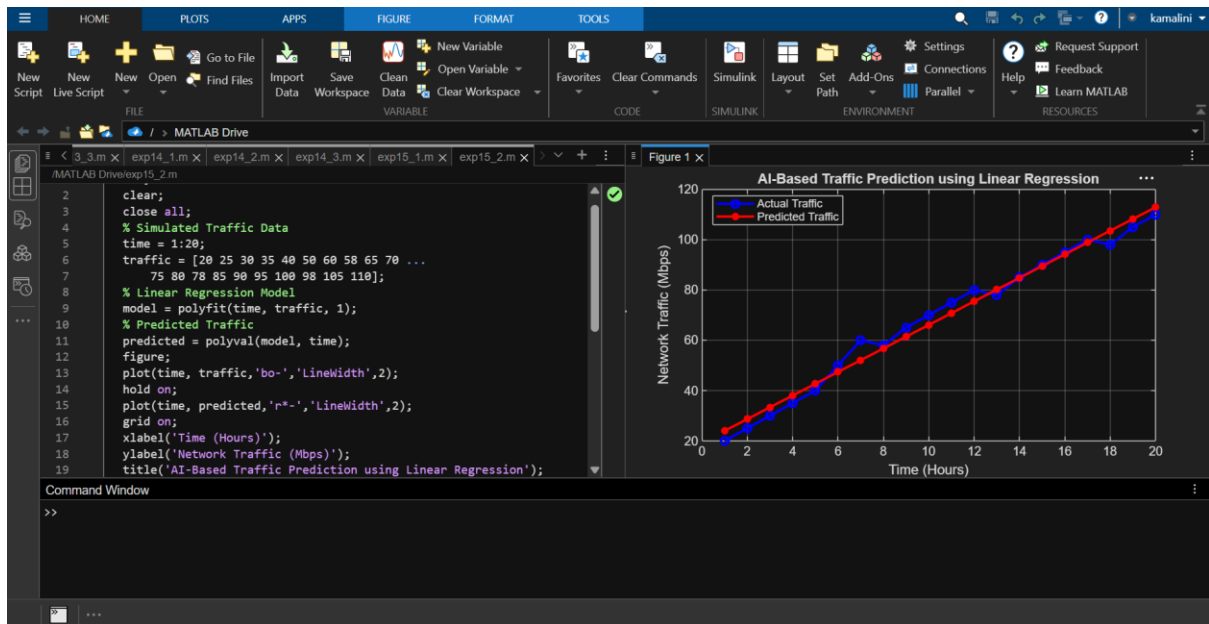
```
clc;
clear;
close all;

% Simulated Traffic Data
time = 1:20;
traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model
model = polyfit(time, traffic, 1);

% Predicted Traffic
predicted = polyval(model, time);

% Plot the results
figure;
plot(time, traffic, 'bo-', 'LineWidth', 2);
hold on;
plot(time, predicted, 'r*- ', 'LineWidth', 2);
grid on;
xlabel('Time (Hours)');
ylabel('Network Traffic (Mbps)');
title('AI-Based Traffic Prediction using Linear Regression');
legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');
```



Objective 3:

To evaluate the prediction accuracy by comparing actual and predicted traffic values.

CODE :

```

clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);

% Predicted Traffic

predicted = polyval(model, time);

% Calculate Prediction Error

predictionError = traffic - predicted;

% Calculate RMSE

RMSE = sqrt(mean(predictionError.^2));

```

```

disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);

% Plot Results

figure;

% ----- Graph 1: Actual vs Predicted -----

subplot(2,1,1);

plot(time, traffic, 'bo-', 'LineWidth', 2);

hold on;

plot(time, predicted, 'r*-', 'LineWidth', 2);

grid on;

xlabel('Time (Hours)');

ylabel('Traffic (Mbps)');

title('Actual vs Predicted Traffic');

legend('Actual', 'Predicted', 'Location', 'best');

% ----- Graph 2: Prediction Error -----

subplot(2,1,2);

bar(time, predictionError);

grid on;

xlabel('Time (Hours)');

ylabel('Prediction Error (Mbps)');

title('Prediction Error');

% Display Table

disp('Time  Actual  Predicted');

disp([time' traffic' predicted']);

```

